Ans. (a) : Number of samples = 150

So probability $= \frac{\text{No. of samples for } (x)}{\text{Total no. of sample } (150)}$

When we consider the given statements

(i) $P(x \le 15) = \frac{3+7+10+15+28+33}{150} = \frac{96}{150} = 0.64$

Hence statement 1 is correct.

(ii) $P(13 < x \le 17) = \frac{28+33+24+11}{150} = \frac{96}{150} = 0.64$

Hence statement 2 is incorrect.

(iii) $P(x = 15) = \frac{33}{150} = 0.22$

Hence statement 3 is also incorrect.

136. At a given time, two players are standing on a play-field. The Cartesian coordinates of their locations are (20, 60) and (-40, –20) units. What is the distance the players?

(a) 60 units (b) 80 units

(c) 100 units (d) 140 units

Ans. (c) : A = (20, 60) and B (– 40, – 20)

$AB = \sqrt{20-(-40)^2+60-(-20)^2}$

$= \sqrt{3600+6400} = \sqrt{10000} = 100$

137. The area of an ellipse is twice that of a circle. The major diameter of the ellipse is twice that of the minor diameter. The radius of the circle is:

(a) 50% of minor diameter of the ellipse

(b) 50% of major diameter of the ellipse

(c) Minor diameter of the ellipse

(d) Major diameter of the ellipse

Ans. (a) : Let the minor diameter of ellipse be $2x$

major diameter of ellipse = 2y which is equal to 2 ($2x$)

$\therefore \quad 2y = 4x$

$y = 2x$ .....(i)

Let the radius of circle be r

Then $\pi \times x \times y = 2\pi r^2$

$\pi \times x \times 2x = 2\pi r^2$

$2\pi x^2 = 2\pi r^2$

$x = r$

$\therefore$ radius of the circle is $\frac{1}{2}$ of the diameter of minor ellipse or 50% of minor diameter of the ellipse.

138. Which one of the following statements is not correct? Dinar/New Dinar is the currency of :

(a) Sudan (b) Yugoslavia

(c) U.A.E (d) Tunisia

Ans. (c) : Currency of Sudan, Yugoslavia and Tunisia is dinar, whereas currency of U.A.E. is Dirham.

139. Which one of the following pairs is not correctly matched?

(a) Jim Laker : Highest number of wickets in a Cricket Test Match.

(b) Brain Lara : Highest score in an innings in Test Cricket.

(c) Sanath Jayasurya and Roshan Mahanama : Highest partnership in a Test Cricket.

(d) Sunil M. Gavaskar : Highest aggregate of runs in Test Cricket.

Ans. (d) : Among the four given option, option (d) is not correctly matched because the highest total of runs in Test cricket which was scored by Sunil Gavaskar was broken by Allan Border of Australia. Sachin Tendulkar is currently the batsman with the highest total of test cricket runs.

140. Which one of the following pairs of folk dance forms and states is not correctly matched?

(a) Korku : Maharashtra

(b) Jhummar : Haryana

(c) Thali : Himachal Pradesh

(d) Mukna : Manipur

Ans. (c) : Thali is the folk dance of Rajasthan, not the folk dance of Himachal Pradesh.

141. In order to win the Grand Slam in Tennis, a player must win which one of the following groups of tournaments?

(a) Australian Open, Wimbledon, French Open, U.S. Open

(b) Wimbledon, French Open, U.S. Open

(c) Wimbledon, French Open, Paegas Czech Open, U.S. Open

(d) Davis Cup, Wimbledon, French Open

Ans. (a) : In order to win the Grand Slam in Tennis, a player has to win four events in a calendar year, which are as follows - Australian Open, French Open, Wimbledon Championship and U.S open.

142. Amar, Akbar and Anthony are friends, being looked after by a Farah, Amar weighs 50% more than Akbar and Anthony weights 25% less than Amar. Farah weighs a third of the combined weight of the three boys. All four together weigh 232 kg. The correct arrangement of the persons in the ascending order of their weights is:

(a) Anthony, Akbar, Farah, Amar

(b) Anthony, Akbar, Amar, Farah

(c) Akbar, Anthony, Amar, Farah

(d) Akbar, Anthony, Farah, Amar

Ans. (d) : Let Akbar's weight be x

$\therefore$ Amar's weight 50% more than $x = \frac{3x}{2}$ .....(i)

And Anthony's 25% less than Amar's $= \frac{3x}{2} - \frac{3x}{8} = \frac{9x}{8}$

.....(ii)

Farah's weight $= \frac{1}{3}\left(x + \frac{3x}{2} + \frac{9x}{8}\right)$ .....(iii)

$= \frac{1}{3}\left[\frac{8x+12x+9x}{8}\right] = \frac{29x}{24}$

$ATQ = x + \frac{3x}{2} + \frac{9x}{8} + \frac{29x}{24} = 232$

$= 116x = 232 \times 24 \Rightarrow x = 48$

Akbar = 48 kg.

Amar = 72 kg.

Anthony = 54 kg. [from equ. (i), (ii) and (iii)]

Farah = 58 kg.

Hence the correct ascending order is Akbar < Anthony < Farah < Amar.

143. In the sequence of numbers 5, 8, 13, X, 34, 55, 89 ..... the value of 'X' is "

(a) 20 (b) 21

(c) 23 (d) 29

Ans. (b) : The given series is formed by adding the previous 2 terms, i.e 8+5=13. Similarly 8+13=21, which further gives 21+13=34, hence B is right answer.

144 On another planet, the local terminology for earth, water, light, air and sky, are 'sky', 'light', 'air', 'water' and 'earth' respectively. If someone is thirsty there, what would he drink"?

(a) Sky (b) Water

(c) Air (d) Light

Ans. (d) : On another planet - soil, water, light, air, sky Local name - sky, light, air, water, soil, so if a person feels thirsty on that planet, he will drink the light.

145. The average speed of a train in the onward journey is 25% more than that of the return journey. The train halts for one hours on reaching the destination. The total time taken for the complete to and fro journey is 17 hours covering a distance of 800 km. The speed of train in the onward journey is:

(a) 45 km per hour

(b) 47.06 km per hour

(c) 50.00 km per hour

(d) 56.25 km per hour

Ans. (d) : Let the speed in return journey be x km/hr.

$\therefore$ speed in onward journey $= x + \frac{25x}{100} = \frac{5x}{4}$ km/hr.

$\therefore$ average speed $= \frac{2 \times \frac{5}{4}x \times x}{\frac{5}{4}x + x}$ km/hr

$= \frac{10x}{9}$ km./hr.

Now time taken to complete to and fro journey minus rest = 16 hr. and distance is 800 km.

Then $800 \times \frac{9}{10x} = 16$

$x = \frac{800 \times 9}{16 \times 10} = 45$

$\therefore$ speed in on ward journey $= \frac{5}{4} \times 45 = 56.25$km/hr.

146. Consider the following statements:

The striking feature of the Jama Masjid in Kashmir completed by Zainul-Aidin include (s)

1. Turret.

2. Similarity with Buddhist pagodas.

3. Persian style

Which of the above statements is/are correct?

(a) 1 alone (b) 1, 2 and 3

(c) 2 and 3 (d) 1 and 3

Ans. (b) : Jama Masjid is a mosque in Srinagar, Jammu & Kashmir. The Jama Masjid of Srinagar is situated at Nowhatta, in the middle of the old city. It was built by Sultan Sikandar in 1400 AD. Later, the son of Sultan Sikandar, Zain-ul-Abidin had extended the mosque. It is similar to buddist Pagoda with Persian style and also includes turret in its architecture.